Q1 (a) BB BG GG GB (b) 1/2

N (Any Boy) = 3

N (One girl in them) = 1

$P = 1/3$

Q3 As, $\text{var}[X] = E[X^2] - (E[X])^2$

$$\text{var}[X+Y] = E[(X+Y)^2] - (E[X+Y])^2$$

$$= E[X^2+Y^2+2XY] - (E[X]+E[Y])^2$$

$$= E[X^2] + E[Y^2] + 2E[XY] - (E[X])^2 - (E[Y])^2 - 2(E[X].E[Y])$$

$$= \left[E[X^2] - (E[X])^2\right] + \left[E[Y^2] - (E[Y])^2\right] + 2\left[E[XY] - E[X]E[Y]\right]$$

$$= \text{var}(X) + \text{var}(Y) + 2\,\text{covar}(X,Y)$$

Q4 $P(P|D) = 99/100$ $\quad P(D) = 1/10000$

$P(P|D') = 1/100$ $\quad P(D') = 9999/10000$

$$P(D|P) = \frac{P(P|D).P(D)}{P(P|D).P(D) + P(P|D').P(D')}$$

$$= \frac{\frac{99}{100} \times \frac{1}{10000}}{\frac{99}{100} \times \frac{1}{10000} + \frac{9999}{10000} \times \frac{1}{100}}$$

$$= \frac{99}{9999+99} \approx 0.0098 = 0.98\%$$

Q5. Contestant chooses door 1 and host opens door 3,

Let C be the rv representing door hiding prize,

$$P(C=1) = P(C=2) = P(C=3) = 1/3$$

Let H be the rv representing host opens door,

$$P(H=3|C=1) = 1/2$$

$$P(H=3|C=2) = 1$$

$$P(H=3|C=3) = 0$$

$$P(H=3) = \frac{1}{2} + \frac{1}{3} + 1 \times \frac{1}{3} + 0 \times \frac{1}{3} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$$P(C=1|H=3) = \frac{P(H=3|C=1).\,P(C=1)}{P(H=3)} = \frac{1/2 \times 1/3}{1/2} = \frac{1}{3}$$

$$P(C=2|H=3) = \frac{P(H=3|C=2).\,P(C=2)}{P(H=3)} = \frac{1 \times 1/3}{1/2} = \frac{2}{3}$$

Hence, $P(C=2|H=3) > P(C=1|H=3)$

$\therefore$ we should switch

Q6 $$P(H|e_1,e_2) = \frac{P(e_1,e_2|H).\,P(H)}{P(e_1,e_2)}$$

If, $e_1 \perp e_2 \,|\, H$, $\quad P(e_1,e_2|H) = P(e_1|H).\,P(e_2|H)$

Hence,

$$P(H|e_1,e_2) = \frac{P(e_1|H) \cdot P(e_2|H) \cdot P(H)}{P(e_1,e_2)}$$

And, $P(e_1,e_2) = \sum_H P(e_1,e_2,H)$

$$= \sum P(H) \cdot P(H|e_1,e_2)$$

$$P(e_1,e_2) = \sum P(H) \cdot P(H|e_1) \cdot P(H|e_2)$$

Q7. Suppose three random binary variables:-

$x_1 \in \{0,1\}$
$x_2 \in \{0,1\}$
$x_3 \in \{x_1 \oplus x_2\}$

x_1	x_2	x_3
0	0	1
0	1	0
1	0	0
1	1	1

For x_1 and x_2,

$P(x_1=1, x_2=1) = \frac{1}{4}$

$P(x_1=1) \cdot P(x_2=1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

For x_2 and x_3,

$P(x_2=1, x_3=1) = \frac{1}{4}$

$P(x_2=1) \cdot P(x_3=1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

For x_3 and x_1,

$P(x_3=1, x_1=1) = \frac{1}{4}$

$P(x_3=1) \cdot P(x_1=1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

∴ pairwise all are independent,

For mutual,

$$P(x_1=1, x_2=1, x_3=1) = \frac{1}{4}$$

$$P(X_1=1) \cdot P(X_2=1) \cdot P(X_3=1) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Hence counter example provided

Q8 We know,

$$X \perp Y | Z \quad \text{iff} \quad p(x,y|z) = p(x|z) \cdot p(y|z)$$

To prove,

$$X \perp Y | Z \quad \text{iff} \quad p(x,y|z) = g(x,z) \cdot h(y,z)$$

$$p(x|z) = \sum_y p(x,y|z)$$

$$= \sum_y g(x,z) \cdot h(y,z)$$

$$= g(x,z) \cdot \sum_y h(y,z)$$

Similarly, $p(y|z) = h(y,z) \cdot \sum_x g(x,z)$

$$\sum_x \sum_y p(x,y|z) = 1$$

$$\sum_x \sum_y g(x,z) \cdot h(y,z) = 1$$

$$\sum_x g(x,z) \sum_y h(y,z) = 1$$

Now, $p(x|z) \cdot p(y|z) = g(x,z) \cdot \sum_y h(y,z) \cdot h(y,z) \cdot \sum_x g(x,z)$

$$= g(x,z) \cdot h(y,z) \cdot \left(\sum_y h(y,z) \sum_x g(x,z) \right)$$

$\therefore \quad p(x|z).p(y|z) = g(x,z).w(y,z)$

Q9. To prove,

$$(X \perp W|Z,Y) \wedge (X \perp Y|Z) \Rightarrow (X \perp Y,W|Z)$$

$$p(y,w|x,z) = p(w|y,x,z).p(y|x,z)$$

For, $(X \perp W|Z,Y) \Rightarrow \quad p(w|y,x,z) = p(w|y,z)$

$(X \perp Y|Z) \Rightarrow \quad p(y|x,z) = p(y|z)$

Hence,

$$(X \perp W|Z,Y) \wedge (X \perp Y|Z) = p(w|y,z)\,p(y|z)$$
$$= p(w,y|z)$$

And,

$$(X \perp Y,W|Z) \Rightarrow p(y,w|x,z) = p(w,y|z)$$

LHS = RHS

Q10 $X \sim Ga(a,b)$

$$y = \frac{1}{x} \qquad \left|\frac{\partial y}{\partial x}\right| = \frac{1}{x^2}$$

$$IG(x|a,b) = G(x|a,b)\left|\frac{\partial G}{\partial IG}\right|$$

$$\Rightarrow IG(x|a,b) = \frac{b^a}{\Gamma(a)}\left(\frac{1}{x}\right)^{a-1} . e^{-b/x} . x^{-2}$$

$$\Rightarrow IG(x|a,b) = \frac{b^a}{\Gamma(a)} \cdot \left(\frac{1}{x}\right)^{a-1} \cdot e^{-b/x} \cdot x^{-2}$$

$$= \frac{b^a}{\Gamma(a)} \cdot (x)^{1-a-2} \cdot e^{-b/x}$$

$$IG(x|a,b) = \frac{b^a}{\Gamma(a)} \cdot (x)^{-(a+1)} \cdot e^{-b/x}$$

Q11

$$I = \int_a^b e^{-x^2/2\sigma^2} dx$$

$$u = \frac{x}{\sqrt{2}\sigma} \Rightarrow du = \frac{1}{\sqrt{2}\sigma} \cdot dx$$

$$\Rightarrow dx = \sqrt{2}\sigma\, du$$

$$I = \sqrt{2}\sigma \int_{-\infty}^{\infty} e^{-u^2} du$$

Now,

$$I = \int_{-\infty}^{\infty} e^{-u^2} du$$

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2} dy\right)$$

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx\, dy$$

$x = r\cos\theta$ $\qquad x^2+y^2 = r^2$

$y = r\sin\theta$ $\qquad dx\,dy = r\,dr\,d\theta$

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} (-r\sin\theta).(r\cos\theta)$$

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} . r\, dr\, d\theta$$

$$I^2 = \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2} dr$$

$$r^2 = K$$
$$2r\,dr = dK$$
$$r\,dr = dK/2$$

$$\frac{1}{2} \int_0^{\infty} e^{-K} dK = -\frac{1}{2} \cdot [e^{-K}]_0^{\infty}$$
$$= -\frac{1}{2} \cdot [0 - 1]$$
$$= \frac{1}{2}$$

$$I^2 = 2\pi . \frac{1}{2}$$
$$\Rightarrow I = (\pi)^{1/2}$$

Hence, $Z = \sqrt{2}\sigma . \sqrt{\pi}$

$$Z = (2\pi\sigma^2)^{1/2}$$

Q12

$$I(X,Y) = \sum_X \sum_Y p(x,y) \cdot \frac{p(x,y)}{p(x) \cdot p(y)}$$

$$= \sum_x \sum_y p(x,y) \cdot \frac{P(x|y) \cdot P(y)}{P(x) \cdot P(y)}$$

$$= \sum_x \sum_y p(x,y) \cdot P(x|y) - \sum_x \sum_y p(x,y) \cdot P(x)$$

$$= -H[X|Y] + H[X]$$

Q13 Now,

$$I(X,Y) = H[X] - H[X|Y]$$

$$I(X,Y) = H[Y] - H[Y|X]$$

$$2I(X,Y) = H[X] + H[Y] - \left[H[X|Y] + H[Y|X]\right]$$

$$-\left(\sum p(x) \cdot \log p(x|y) + \sum p(x) \cdot \log p(y|x)\right)$$

$$-\left(\sum p(x) \cdot \log \frac{P(x|y) \cdot P(y)}{P(y)} + \sum p(x) \cdot \log \frac{P(y|x) \cdot P(x)}{P(x)}\right)$$

$$-\left(\sum p(x) \log p(x,y) - \sum p(x) \cdot \log p(y) + \sum p(x) \cdot \log p(x,y) - \sum p(x) \cdot \log \cdot p(x)\right)$$

$$= H[X,Y] - H[X] + H[X,Y] - H[Y]$$

$$2I(X,Y) = 2H[X] + 2H[Y] - 2H[X,Y]$$

$$I(X,Y) = H[X] + H[Y] - H[X,Y]$$

Q13 We just derived,

$$I(X,Y) = H[X] + H[Y] - H[X,Y]$$

$$I(X_1,X_2) = H[X_1] + H[X_2] - H[X_1,X_2]$$

$$H[X_1] = \frac{1}{2} \cdot \log[(2\pi e\sigma^2)]$$

$$H[X_2] = \frac{1}{2} \log[(2\pi e\sigma^2)]$$

$$\begin{vmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{vmatrix} = \sigma^2 \cdot \sigma^2 - \rho\sigma^2 \cdot \rho\sigma^2$$

$$= \sigma^4 - \rho^2\sigma^4$$

$$= \sigma^4(1-\rho^2)$$

$$H[X_1,X_2] = \frac{1}{2} \log[(2\pi e)^2 \sigma^4(1-\rho^2)]$$

$$I(X_1,X_2) = \frac{1}{2} \log[(2\pi e\sigma^2)] + \frac{1}{2} \log[(2\pi e\sigma^2)] - \frac{1}{2} \log[(2\pi e)^2\sigma^4(1-\rho^2)]$$

$$= \frac{1}{2} \log[(2\pi e)^2\sigma^4] - \frac{1}{2} \log[(2\pi e)^2\sigma^4(1-\rho^2)]$$

$$= \frac{1}{2} \log\left[\frac{(2\pi e)^2\sigma^4}{(2\pi e)^2\sigma^4 \cdot (1-\rho^2)}\right]$$

$$I(X_1,X_2) = -\frac{1}{2} \log(1-\rho^2)$$

For $\rho = 0$, $I(X_1, X_2) = -\frac{1}{2}\log(1) = 0$

$\rho = 1$, $-\frac{1}{2}\log(0) = \infty$

$\rho = -1$, $-\frac{1}{2}\log(0) = \infty$

Q14 (a) $\eta = 1 - \frac{H(Y|X)}{H(X)}$

$= \frac{H(X) - H(Y|X)}{H(X)}$

$= \frac{H(Y) - H(Y|X)}{H(X)}$ Since $H(Y) = H(X)$

$= \frac{I(X,Y)}{H(X)}$

(b) Since, $I(X,Y) \geq 0$ $\quad \frac{I(X,Y)}{H(X)} \geq 0$

$H(X) \geq 0$

$I(X,Y) \leq \min(H(X), H(Y))$

$\leq H(X)$

$\frac{I(X,Y)}{H(X)} \leq \frac{H(X)}{H(X)} = 1$

(c) When X and Y are independent $(I(X,Y) = 0)$

(d) When $I(X,Y) = H(X)$ or say $H(X|Y) = 0$

Q16 Mode :-

$$f(x;\alpha,\beta) = \frac{x^{\alpha-1}\cdot(1-x)^{\beta-1}}{B(\alpha,\beta)}$$

$$\frac{\partial f}{\partial x} = \frac{(\alpha-1)\,x^{\alpha-2}\cdot(1-x)^{\beta-1}}{B(\alpha,\beta)} + \frac{(1-\beta)(x^{\alpha-1})\cdot(1-x)^{\beta-2}}{B(\alpha,\beta)}$$

$$\Rightarrow (\beta-1)(x^{\alpha-1})\cdot(1-x)^{\beta-2} = (\alpha-1)\cdot x^{\alpha-2}\cdot(1-x)^{\beta-1}$$

$$\Rightarrow (\beta-1)x\cdot = (\alpha-1)(1-x)$$

$$\Rightarrow (\beta-1)x = (\alpha-1) - (\alpha-1)x$$

$$\Rightarrow (\beta-1)x + (\alpha-1)x = (\alpha-1)$$

$$\Rightarrow x[\alpha+\beta-2] = (\alpha-1)$$

$$\Rightarrow x = \frac{(\alpha-1)}{(\alpha+\beta-2)}$$

Now,

$$E[x^N] = \int_{-\infty}^{\infty} \frac{x^N\cdot x^{\alpha-1}\cdot(1-x)^{\beta-1}}{B(\alpha,\beta)}$$

$$\frac{1}{B(\alpha,\beta)} \int_{-\infty}^{\infty} x^{(\alpha+N)-1}\cdot(1-x)^{\beta-1}\,dx$$

$$E[x^N] = \frac{B(\alpha+N,\beta)}{B(\alpha,\beta)}$$

$$E[x^N] = \frac{\frac{\Gamma(\alpha+N)\,\Gamma(\beta)}{\Gamma(\alpha+N+\beta)}}{\frac{\Gamma(\alpha)\;\Gamma(\beta)}{\Gamma(\alpha+\beta)}}$$

$$= \frac{\Gamma(\alpha+N)}{\Gamma(\alpha+\beta+N)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)}$$

$$= \frac{(\alpha+N-1)!}{(\alpha+\beta+N-1)!} \cdot \frac{(\alpha+\beta-1)!}{(\alpha-1)!}$$

For $n=1$,

$$\frac{(\alpha)!}{(\alpha+\beta)!} \cdot \frac{(\alpha+\beta-1)!}{(\alpha-1)!}$$

$$= \frac{\alpha}{\alpha+\beta}$$

For $n=2$,

$$\frac{(\alpha+1)!}{(\alpha+\beta+1)!} \cdot \frac{(\alpha+\beta-1)!}{(\alpha-1)!}$$

$$= \frac{(\alpha)\,(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$$

$$\text{Mean} = E[x] = \frac{\alpha}{\alpha+\beta}$$

$$\text{Variance} = E[x^2] - (E[x])^2$$

$$= \frac{(\alpha)\,(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \frac{(\alpha^2)}{(\alpha+\beta)^2}$$

$$= \frac{(\alpha)(\alpha+1)(\alpha+\beta) - (\alpha)^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{(\alpha^2+\alpha)(\alpha+\beta) - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{\alpha^3+\alpha^2\beta+\alpha^2+\alpha\beta - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$